- # Stakeholder Analysis and Conflict Resolution

Stakeholder Analysis

Method: Stakeholders were identified from the case study and the five interview statements (Principal, Lecturer, Head of Department, Student Representative and IT Officer). Each stakeholder was then rated for:

1. Influence, which is how much the stakeholders affect what can be built or how the system must operate.
2. Interest, which is how much the system affects the stakeholders

The analyzing criteria used is HIGH, MEDIUM and LOW

Table 4: Stakeholders and their interests

	Stakeholder	Role in the system	Main interests	Influence	Interest
1	Principal	Sponsor and final decision maker	A simple system for non-IT staff. Students must not see any mark until the Head of Department has checked it.	High	Medium
2	Head of Department (HoD)	Reviews, moderates and releases marks	Sees every mark in the department before release. Late-penalty rule applied consistently (10% per day for up to three days, then zero).	High	High
3	Lecturers	Create assignments, grant extensions, mark work, write feedback	Students get marks soon after marking. Ability to grant extensions for good reasons. Easy to use.	Medium	High

	Stakeholder	Role in the system	Main interests	Influence	Interest
4	Students (via student representative)	Submit work, resubmit, view released marks and feedback	Fix a mistake and resubmit before the deadline. Written feedback, not just a number. Late work not simply rejected.	Low	High
5	System administrator	Manages users, departments, courses, assigns lecturers, enrols students	Simple account and course management. Reliable system.	Medium	High
6	IT officer	Installs and maintains the system	Works offline on old computers. Passwords protected. Fast response with 800 students stored.	Medium	Medium
7	Parents and guardians (indirect)	Not users	Learners' personal data and marks kept private and accurate.	Low	Medium
8	Project supervisor	Reviews each phase gate	Requirements that are complete, consistent and verifiable.	High	Medium

Table 5: Influence and interest grid

	Lower interest	Higher interest
Higher influence	**Keep satisfied:** Principal, Supervisor	**Manage closely:** Head of Department
Lower influence	**Monitor:** Parents and guardians	**Keep informed:** Lecturers, Students, Administrator, IT officer

How the grid is used. The Principal and HoD decide the rules that most affect the design, so their statements carry the most weight when conflicts are resolved. Lecturers, students and the administrator use the system every day, so their needs shape the interface and workflow. Parents are not users, but they are the reason data protection and privacy are treated as requirements.

Ambiguities and Resolutions

Resolution rules. When stakeholders disagreed, or a statement was unclear, I applied these rules in order:

1. Rules stated in the case study come first.
2. Legal and ethical duties come next, such as protecting learners' data and treating them fairly.
3. The Principal's decision comes after that, as the person in charge of the school, while still meeting the other stakeholder's real need in another way wherever possible.

Table 6: Conflicts between stakeholders

	Conflict	Resolution	Reason
1	**When students see marks:** The Principal says not until the HoD has checked them. The Lecturer wants students to see them as soon as they are marked.	Students see marks and feedback only after the HoD has moderated and released them. The lecturer's need for speed is met by notifying the HoD as soon as marking is finished.	The case study makes the HoD responsible for releasing marks, and the Principal has the final say.
2	**Whether late work should be accepted or rejected:** The student representative says late work must not be rejected. The HoD says it loses marks and then gets zero.	Late work is always accepted and recorded. A penalty is applied, and after three days late the mark is zero. The lecturer can still write feedback.	Meets both needs: the work is never rejected, and the HoD's penalty rule still applies.
3	**Extensions versus strict penalties:** The Lecturer wants to grant extensions. The HoD wants late work to be penalized.	A lecturer can grant an individual extension with a written reason, recorded in the system. The penalty is counted from the new deadline.	Fair to students with a good reason, and every extension is traceable so the HoD can review it.

	Conflict	Resolution	Reason
4	**Simple system versus security and audit:** The Principal wants simplicity. The IT officer wants protected passwords, and the case study requires a log of every mark change.	Security works in the background (salted password hashes, audit log). The interface stays menu-driven and shows only the options for the user's role.	Users do not see the complexity, so both needs are met.

Table 7: Ambiguities and incomplete statements

	Ambiguity	Resolution
1	"10% of the available mark for each day late": 10% of what, and what counts as a day?	The penalty is 10% of the assignment's maximum mark for each day or part of a day late, taken off the earned mark and never going below zero. More than three days late gives zero as last mark. Example: maximum 20, earned 16, two days late: penalty 4, final mark 12.
2	Lateness is measured from which deadline?	From the effective deadline: the extended deadline if one was granted, otherwise the original deadline.
3	"Resubmit before the deadline": what happens to earlier versions?	Resubmission is allowed only before the effective deadline. The latest version is the one marked, and earlier versions are kept in the submission history.
4	What does the HoD actually do?	The HoD reviews every mark in their department, may change a mark with a reason (recorded in the audit log), then approves and releases it.
5	"Respond quickly" is not measurable.	Replaced by measurable targets in the SRS, for example login within 2 seconds with 800 learner records stored.

	Ambiguity	Resolution
6	"Single computer or local network": SQLite is a single file and is unreliable when several computers write to it over a network share.	**The system runs on one school computer.** The program and the SQLite database file are stored on that computer, and users work on it directly. The database file is never shared over the network. All database access goes through the data-access layer, so a networked version could be added later by changing only that layer. To be confirmed with the IT officer.
7	When do students see feedback?	At the same time as the marks, after HoD release, to stay consistent with conflict 1.
8	Until when can an extension be granted?	Until the submission has been marked. If a student has already submitted late, the penalty is recalculated against the new deadline.
9	Can a mark change after it has been released?	Yes, but only the HoD can change a released mark, a reason is required, the change is recorded in the audit log (who, when, old value, new value, reason), and the student sees the updated mark.

Table 8: Business rules that result from the resolutions

These rules will become requirements in the SRS and will be used in the design, code and tests.

Rule	Statement	From
BR1	A student can see a mark and feedback only after the HoD has released it.	C1, A7
BR2	Late work is always accepted. Penalty = 10% of the maximum mark per day or part of a day late, up to three days. After three days the mark is zero.	C2, A1

Rule	Statement	From
BR3	Lateness is measured from the effective deadline, which includes any extension.	A2
BR4	An extension needs a written reason and is recorded. It can be granted until the submission is marked.	C3, A8
BR5	A student may resubmit only before the effective deadline. The latest version is marked and earlier versions are kept.	A3
BR6	The HoD can review, change (with a reason) and release all marks in their department.	A4
BR7	Every mark change is recorded with who, when, old value, new value and reason. Only the HoD can change a released mark.	C4, A9
BR8	The system runs on one school computer with a local SQLite database.	A6